

Review

Expert systems: Definitions, advantages and issues in medical field applications

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ABSTRACT

The aim of this review is to provide a broad overview of the state-of-the-art works mainly published in the last ten years on expert systems applied in different medical domains.

Being able to support and sometimes substitute experts, an expert system may be a precious ally for medical diagnoses. Medical expert system applications provide physicians and patients with an immediate access to knowledge and advice, rooting their flexibility into their knowledge bases, rule sets and graphical interfaces. To be trusted by their users, medical expert systems should follow some criteria, which we investigate along with their different realization, from fuzzy logic to wearable solutions for out-of-clinical-environment care. We also consider the advantages of approaching diagnoses and alert systems through an artificial intelligence counterpart, without forgetting the importance of a good validation to assess the system functionality.

Therefore, we show the heterogeneity of the solutions proposed by the literature, bounded to the specific needs a medical expert system is called to answer, the common lack of a system validation and the possible benefits deriving from these systems application.

1. Introduction

Expert systems (ESs) are a mean through which share and distribute knowledge (Hendriks, 1999; Oleshchuk & Fensli, 2011), being it acquired directly or indirectly from domain experts of different scientific areas. They are not only an aid to users lacking a specific know-how, but also supports (Oyediji, Osifeko, Folorunsho, Abolade, & Ade-Ikuesan, 2019), if not even substitutes (Ahmed et al., 2019), for human experts. Therefore, the power of an ES should lay in its ability to mimic the human decision process (Kim, Oh, & Pedrycz, 2017; Mirmozaffari, 2019), especially when applied in the medical field.

In fact, many applications have been developed to support the diagnoses of different illnesses, suggest treatment procedures and/or monitoring of patients' conditions (Abu-Nasser, 2017), not only in a clinical environment, but also at home (Mahajan & Kaur, 2016; Machado, Maran, Augustin, Wives, & de Oliveira, 2017).

The synergy of ES inferences with physicians and human experts has proved to bring improvement to the diagnosis and treatment processes, both in terms of reliability and efficiency (Singla, Grover, & Bhandari, 2014; Abu-Nasser, 2017; Rivera, Pagaduan, Caliwag, Reyes, & Castillo, 2019).

Taking into consideration the great variety of application domains an expert system could work on, each model has assumed peculiar characteristics suitable to its problem of interest (Nohria, 2015; Josefiok, Krahn, & Sauer, 2017). As further development of ESs we analyze: (1) fuzzy expert systems (FESs), whose knowledge is based on vagueness in order to be more versatile in facing complex problems (Sikchi, Sikchi, & Ali, 2013; Viharos & Kis, 2015; Rajabi, Hossani, & Dehghani, 2019); (2) medical expert systems (MESSs), the core of our review, especially made to assist in the diagnosis (Drweesh & Al-Bakry, 2019) and care processes and which should be able to handle the uncertainty typical of symptoms analysis by relying on well-defined knowledge bases (Nohria, 2015) and eventually on the fuzzy logic of FESs (Dev, Sultana, & Mitra, 2016); (3) wearable expert systems (WESs), which use continuous data collection from different sensors to monitor a subject for a long period of time, also remaining connected to their knowledge base and being able to update it at any moment (Sartori & Melen, 2017).

Moreover, our review focuses on analyzing ESs with special attention to their advantages and issues (Duan, Edwards, & Xu, 2005; Sheikhtaheri, Sadoughi, & Dehaghi, 2014), which should also be indexes to understand if the modeling of an ES is going to represent a real aid to solve peculiar issues. Also, since ESs have specific functioning and

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architecture, they bring unique problems that need to be addressed correctly in order to allow a real benefit to their users (Asabere, 2012; Santra, Mandal, Basu, & Goswami, 2019).

We then highlight the importance of a validation step as a mean through which understand if a proposed system is efficient, accurate and reliable, being it used in un-/supervised real-life situations (Asabere, 2012; Maylawati, Darmalaksana, & Ramdhani, 2018).

Knowing all the characteristics a MES should retain in order to be trusted, we analyze some state-of-the-art works mainly concerning heart, neurological, pulmonary and urinary system diseases, elderly healthcare and fall detection applications.

In order to explore the large domain of expert systems, we made a search on Google Scholar (<https://scholar.google.com/>) using some relevant combined keywords such as (sorted in alphabetical order): *advantages, benefits, challenges, disease, expert systems, fuzzy, healthcare, inference engine, issues, knowledge base, knowledge-based, medical, user interface, solution, technologies, validation, wearable*.

The initial paper selection was performed by screening titles and abstracts, then we selected the articles that provide a concrete example of an ES application in the healthcare domain, in respect to the following criteria:

- The article is related to expert systems or knowledge-based solutions;
- The work includes the description of at least one of the following ES constituting part: knowledge base, inference engine, user interface;
- The article date of publication is between 2008 and 2020 and the ones before 2010 have been cited after 2010.

Notice that we mainly considered the works published in the last 5 years. The overall paper distribution over time is reported in Fig. 1.

Moreover, the selected papers belong mainly to well-known publishers, such as IEEE, Elsevier, Springer (Fig. 2), and we divided them according to their specific application in the medical field (Fig. 3).

Summarizing, we identified 43 articles that are relevant for our study and in particular for the section related to the MESs applications (Section 5).

Our paper is organized as follows: in Section 2, we describe the basic concepts underlying the ESs and we detail some aspects regarding the ES based on mathematical tools able to make a decision in presence of uncertainty (Section 2.1), the ES modeled for medical applications (Section 2.2) and the ES based on wearable healthcare technologies (Section 2.3), in order to make our point on the advantages deriving from the integration of these systems.

In Section 3, we describe advantages and issues deriving from the modeling of a generic ES, while in Section 4, we highlight the importance of including a validation step during the system deployment.

We then report some MES applications (Section 5), basing our subdivision on the various paper topics of interest, i.e., heart-related applications, fall detection and elderly problems, neurology and chronic

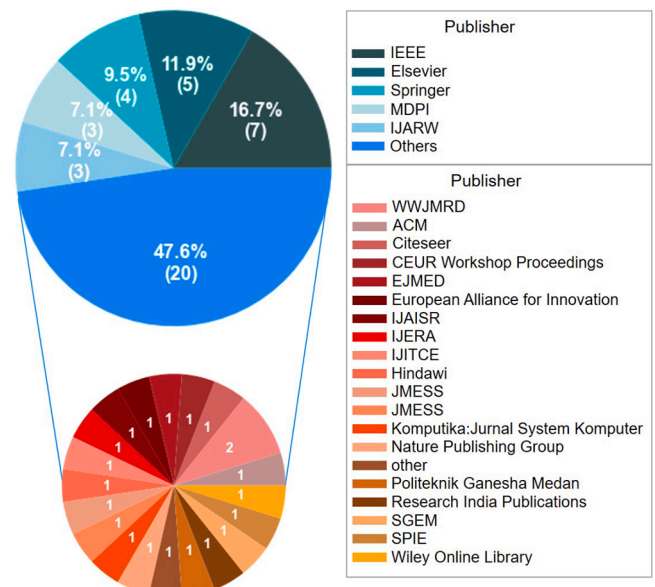


Fig. 2. Paper distribution by publisher.

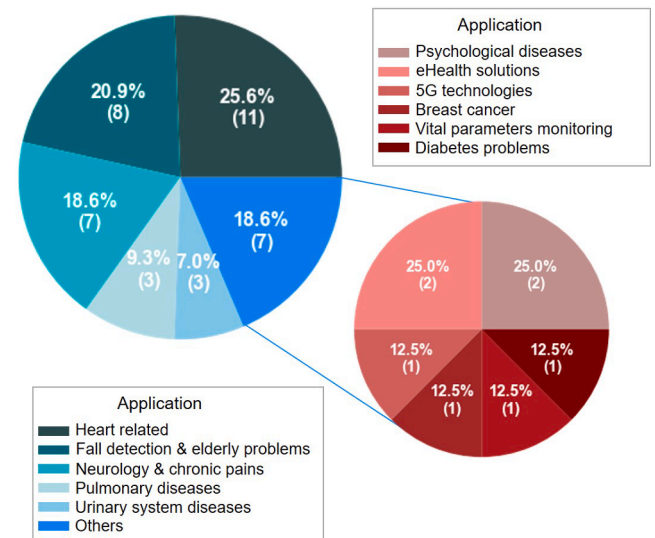


Fig. 3. Paper distribution by medical application.

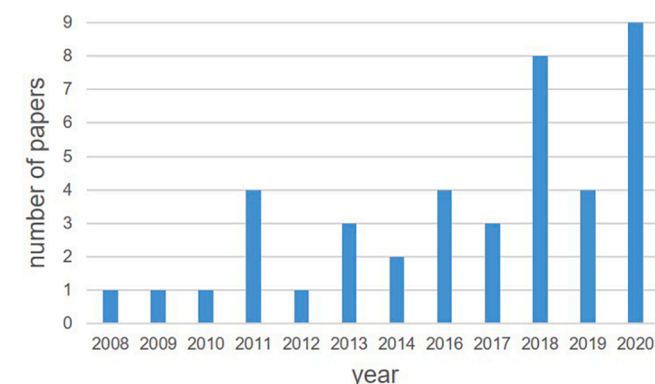


Fig. 1. Paper distribution by publication year.

pains, pulmonary and urinary system diseases and other medical applications.

In Section 6 we discuss some critical aspects coming from our study and we conclude our work by giving some final remarks (Section 7).

2. Expert systems

Starting from the mid 60s, where the first ESs were developed (Liao, 2005), the definition of *expert system* has evolved with the advances of Information and Communications Technology (ICT) solutions (Das, Ghosh, & Kar, 2013; Singla et al., 2014; Abu-Naser & Shaath, 2016; Abu-Naser & AlDahdooh, 2016; Vourgidis, Mafuma, Wilson, Carter, & Cosma, 2018; Oyediji et al., 2019; Tavana & Hajipour, 2019), while maintaining its characteristic core:

1. It is a knowledge-based system, meaning that it is provided with a database (Oyediji et al., 2019) containing and, constantly, collecting

- all the information needed to access and address a peculiar problem (Ahmed et al., 2019);
2. It emulates the decision-making process performed by a human expert (Mirmozaffari, 2019), being mainly provided with an inference engine that is able to process a rule set based on the collected knowledge (Ahmed et al., 2019);
 3. It is an application of Artificial Intelligence (AI) (Bahrammirzaee, 2010; Liao, 2005), which provides technical solutions (Ruiz-Mezcua, Garcia-Crespo, Lopez-Cuadrado, & Gonzalez-Carrasco, 2011) for data collection and system behaviour modeling.

In the 80s, Shortliffe (1986) and Lachman (1989) described this core from another perspective, i.e., from a physician centric perspective and from a cognitive science one, respectively. An ES is anyhow depicted as a system based on the explicit knowledge derived from experts in different and autonomous fields (Shortliffe, 1986). This knowledge could be structured as rules, without the rigidity of conventional programs (Shortliffe, 1986; Lachman, 1989). Therefore, the ambiguity is not an issue for these ESs, having the intelligence as part of the system (Lachman, 1989).

The knowledge is decoupled from the context, thus having easier to understand and modify data (Lachman, 1989). However, the ESs become unpredictable due to the dependence from the semantic routines for the system control, which could follow a path not known a priori (Lachman, 1989).

In more recent advances in the expert system development and application (Das et al., 2013; Singla et al., 2014; Abu-Naser & Shaath, 2016; Abu-Naser & Aldahdooh, 2016; Vourgidis et al., 2018; Oyediji et al., 2019; Tavana & Hajipour, 2019), the definition acquired a number of technical details, maintaining its 3 core characteristics.

Firstly, necessary modules constituting an ES (Abu-Naser & Aldahdooh, 2016) are identified, i.e., a user interface to allow a better access to knowledge and the possible deriving advice (Das et al., 2013); an inference engine to explicate the decision-making process that has been taken and then an explanation subsystem to provide information about the inference and reasoning processes underlying the ES to the user (Tavana & Hajipour, 2019); a knowledge base storing both the information derived from experience and the rules for its application (Singla et al., 2014).

Secondly, Vourgidis et al. (2018) recently introduced the concept of trust in ESs, having analyzed their application in the medical field. While we discuss in more depth the concept of trust in Section 2.2, we find that the authors (Vourgidis et al., 2018) identified 3 characteristics that should be present in every ES: (1) consistency, meant as the ability to obtain the same results from the same set of inputs; (2) availability, meaning that the knowledge of the ES is always present and is a built-in that does not require years of training as a human expert; (3) comprehensiveness, meant as the ability to collect the knowledge derived from different human experts and thus give different pertinent outputs.

Lastly, the ESs began to be seen not only as an aid to human experts, but also as substitutes (Ahmed et al., 2019) for a decision making process. However, notice that the simulations are designed by a specific person able to model the human behavior and translate it into machine-understandable rules (Abu-Naser & Shaath, 2016).

Another aspect of the expert systems is that their knowledge bases could have different representations, being them more rule-based or AI-techniques driven (Josefiok et al., 2017). This diversity allows the application of an ES in different fields like healthcare, engineering and education support (Oyediji et al., 2019).

In this work we focus our analysis on the medical expert systems (MESs), even in the case of mobile and wearable device applications.

Therefore, in the following we divide our theoretical analysis in 3 sections, wanting to (1) explicate the power of fuzzy expert systems and other mathematical models in handling the uncertain data typical of the medical field (Section 2.1); (2) give more insight about the MESs as efficient and reliable tools for assisting both physicians and patients

(Section 2.2); (3) highlight the usefulness of wearable healthcare technologies, wanting to have a constant monitoring and assistance of the patients even in an out of the hospital environment (Section 2.3).

2.1. Fuzzy expert systems

Before introducing the MESs, we briefly describe fuzzy expert systems (FESs) as an evolution of ESs used by some of the cited authors, mainly to handle the uncertainty of medical knowledge bases.

Notice that the fuzzy logic is not the only mathematical method able to address the problem of uncertainty, but that we focus on it having found that most of the reported (Section 5) literature works use fuzzy logic extensively.

Nevertheless, we describe some alternative methods for completeness in the final paragraphs of this section.

Being in presence of ambiguity, the core of a FES is fuzzy logic, which gives good reasoning mimicking the human decision process and follows a knowledge representation based on membership (Rajabi et al., 2019).

Thus, the FES receives an input in the form of continuous numbers and translates them into linguistic terms (Nohria, 2015), i.e., fuzzy variables. This process is called *fuzzification*, where the linguistic terms and the corresponding membership functions are defined by collecting information from human experts, the data and literature from the field of interest (Das et al., 2013). Afterwards, the fuzzy inference engine provides the rules in order to map the fuzzy variables on similar output fuzzy variables (Nohria, 2015). As a final step the *defuzzification* translates the linguistic terms back to numbers (Kadhim, Alam, & Kaur, 2011).

Therefore, FESs are able to define knowledge based on vagueness and expertise, being medical diagnosis expressed in natural language and by different medical experts (Sikchi et al., 2013). They give linguistic concepts that could be exploited for early diagnosis and could also represent a mean through which allow an immediate communication and knowledge sharing between physicians, patients and developers (Sikchi et al., 2013).

As a final remark on FESs notice that, in recent works (Abushariah, Alqudah, Adwan, & Yousef, 2014; Schipor, Geman, Chiuchisan, & Covasa, 2016; Shihabudheen & Pillai, 2018), they have been combined with artificial neural networks (ANNs) to obtain systems (1) able to learn and tune parameters and (2) to use a fuzzy logic, thus versatile to solve complex problems (Viharos & Kis, 2015).

As anticipated, we conclude this section by introducing some of the other mathematical methods that are able to model vagueness and uncertainty.

Pawlak's rough set theory (Pawlak, 1982) provides an exact mathematical model to induce concept approximations, without requiring preliminary or additional information about the data we want to analyze (Ilczuk, Mlynarski, Wakulicz-Deja, Drzewiecka, & Kargul, 2005). It allows data reduction by finding the minimal sets retaining the same knowledge, detects dependencies, evaluates data significance, it is able to generate decision rules, discovers patterns and cause-effect relationships in the data (Pawlak, Grzymala-Busse, Slowinski, & Ziarko, 1995).

Exploiting these characteristics, Ilczuk et al. (2005) develop a rule induction algorithm based on rough set theory and obtain an attribute reduction and a decision rule generation with an increase in accuracy and a decrease of the computation time, thus they enable rule testability and verification by medical experts.

Another approach is based on Dempster-Shafer theory, also called theory of belief functions (Shafer, 1992), or on its variations. In fact, associating a rule with a belief degree is able to better model continuous causal relationships in respect to if-then rules (Sevastianov, Dymova, & Bartosiewicz, 2012) and thus may represent a good solution for medical decision support systems.

A development of belief based theory is the rule-base inference methodology using the evidential reasoning (RIMER) (Yang, Liu, Wang, Sii, & Wang, 2006): the rule base follows a scheme in which belief

degrees are embedded in the rule consequents, providing a system with the ability to model uncertainty. Moreover, the inference relies on the evidential reasoning approach in order to combine rules and come to useful conclusions (Yang et al., 2006).

However, Sevastianov et al. (2012) highlight 2 main issues of RIMER: the belief degree is not associated to a group of hypotheses, but to a specific hypothesis, and does not provide a combination of sources, that are typical of the medical domain.

Apart from these approaches, there are many solutions in the soft computing field, whose techniques are adaptable to a specific problem domain and thus exploitable in the healthcare domain (Gambhir, Malik, & Kumar, 2016). However, we do not discuss them in the present work, being more interested in the specific applications of medical expert systems.

2.2. Medical expert systems

Medical expert systems are born from the necessity of assisting physician and patients in the diagnosis and treatment of diseases, even as a mean through which make alerts and notify physicians and patients (Almarashdeh et al., 2018). They could also represent a change for better in the quality of life, being always able to assist their users (Seto et al., 2012).

The synergy of information technology and clinical knowledge is then intended to improve healthcare services, in terms of safety, efficiency and reliability (Silva, Rodrigues, de la Torre Díez, López-Coronado, & Saleem, 2015). Therefore, being healthcare a sensitive and evolving field of application, these expert systems need to be continuously updated, expanding or modifying their knowledge base and the rule set (Nohria, 2015).

To obtain this update, Nohria (2015) identified 2 phases: (1) the detailed description of the diseases and their symptoms are collected exploiting the physicians' knowledge and the patients' feedback; (2) symptoms and diseases are modeled as rules, where the firsts are antecedents and the seconds consequences.

Knowing this, the MESs have to handle knowledge that could be uncertain and susceptible to noise (e.g. biomedical data) and thus require the development of a suitable model (Das et al., 2013).

Having such a level of uncertainty, that is mainly based on human experience and that could not be controlled as a single well-defined rule, the medical knowledge could be modeled as a fuzzy relation (Dev et al., 2016), even exploiting the previously described fuzzy expert systems.

Finally, the development of a MES needs to take into consideration the concept of trust (Vourgidis et al., 2018). In healthcare, trust could be considered a human relationship that involves both a cognitive component and the efficacy of treatment, diagnosis, communication between the patient and the physician (Vourgidis et al., 2018). Translated into the expert systems, trust could be represented by (1) the user's level of acceptance, being it referred to the validity and quality of the information given by the system, and (2) the reliability of the system itself (Vourgidis et al., 2018).

Trust should also be a concern when a MES relies on mobile and wearable devices to collect data from and connect to the patients.

We discuss in Section 2.3 the main characteristics of wearable expert systems.

2.3. Wearable expert systems

Wanting to take health monitoring out from a clinical environment and to allow a continuous data collection over long periods of time, the need of wearable systems arises (Bonato, 2003).

The idea underlying this need is to give patients non-invasive devices that are able to record physiological data with a good resolution (even in presence of noise) and to send them to clinical databases (Bonato, 2003; Yetisen, Martinez-Hurtado, Ünal, Khademhosseini, & Butt, 2018). This transmission could also represent a way to generate alerts in case the

collected parameters result abnormal following specific criteria (Chowdhury et al., 2019).

A wearable expert system (WES) could be defined as an ES which, given a time varying input of data collected by wearable devices, outputs dynamic responses to the same wearable devices (Sartori & Melen, 2019).

According to Sartori and Melen (2017) a WES should have an agile development cycle, be scalable and have a continuous update of its knowledge base (depending on the advancement in the healthcare field and on the variation of the users' recorded data), be always connected to a centralized knowledge maintenance system.

This kind of ESs should also use a proper set of features to address the problem of interest and consider the constraints deriving from the use of mobile technologies and the subjectivity of the collected data (Rescio, Leone, & Siciliano, 2013).

Concerning the mobile technologies, in the last decade 5G technologies along with IoT (Internet of Things), the collection of a huge amount of data and the advancements in artificial intelligence (Latif, Qadir, Farooq, & Imran, 2017) have been identified as boosters to telemedicine and eHealthcare.

Even though 5G was not a mature technology in 2011, Oleshchuk and Fensli (2011) described its potential to allow new solutions for patient monitoring, being the systems able to connect to experts in case of critical conditions and to continuously share information.

The appeal of 5G technology comes especially from its characteristics: low latency and energy consumption, high system capacity, connectivity and security (Chowdhury, Hossan, Shahjalal, Hasan, & Jang, 2020; Ullah et al., 2019).

3. Advantages and issues

Another contribution highlighted by Lachman (1989) in the late 80s (Section 2) is related to the explanation of the advantages brought by the introduction of ESs in the cognitive sciences in respect to other traditional applications. One of his main concerns was to emphasize the limitation of conventional programs in peculiar applications.

While they often lead to a high computational and time complexity, especially when facing problems in possibly ill defined, open-ended and multi-disciplinary fields, expert systems are able to limit the search space and the problem solutions by relying on heuristic methods based on facts, opinions and information from a domain expert (Lachman, 1989). As introduced in Section 2, this is possible knowing the characteristics of an ES: the representation of knowledge and the decision process mimic the human experts' know-how and behavior, thus avoiding the limitation given by a static problem structure.

Another major difference between conventional software and ESs lays in where their capabilities are stored (Lachman, 1989). While the former are largely defined by the implemented algorithms and procedures, the latter find their *intelligence* in their coded knowledge, that can be represented in different ways depending on the expert system (e.g. rules, semantic networks, frames) and that can then be processed by an inference engine (Lachman, 1989; Duan et al., 2005). Since the knowledge of an ES is self contained in its knowledge base and its rules, the intelligence behind expert systems is context-independent, easy to modify and easier to understand compared to the one behind conventional programs (Lachman, 1989). Even though these are appealing characteristics, they also bring some issues: ESs could be unpredictable in their behavior and responses, thus making their routines and the consequences of their processing more difficult to anticipate and control (Lachman, 1989; Marlinda, Widiyawati, Widiastuti, & Indrarti, 2020).

Over the late 80s, Thomas (1989) extended the analysis on the intrinsic characteristics of the ESs, deeply investigating on the benefits and issues they brought.

The highlighted advantages of the ESs can be summarized in the following way: (1) ESs can provide expertise in a more convenient way, being more economic and accessible than their human counterpart, and

thus contributing in the spreading of expert knowledge to broader users (Thomas, 1989; Sheikhtaheri et al., 2014); (2) ESs can relieve human experts from some tasks, lightening their workload and allowing them to focus on more demanding analyses (e.g. by performing routine diagnosis on patients' symptoms or analysis on control information) (Thomas, 1989; Asabere, 2012); (3) ESs can provide a more accessible and useful way for human experts to develop and test new theories, especially in the healthcare field (Thomas, 1989).

On this last point, notice that being able to store large quantities of data and to reason on increasingly more complex information, ESs provide support to the human experts by reasoning in parallel with them (Ahmed et al., 2019; Oyediji et al., 2019).

On the other hand, Thomas (1989) described the issues brought by ESs, maybe explaining why they are not as widely used as their benefits would suggest.

Firstly, expert systems require a specific shell (i.e., a program), which could be difficult to operate for the final user, and a carefully organized knowledge base to function properly (Thomas, 1989; Sheikhtaheri et al., 2014). This usually requires the intervention of experienced programmers or even a full development team in order to correctly model the ES, raising both production times and costs (Thomas, 1989).

Secondly, ESs offer a very restrictive way of storing knowledge and adding new information to their initial knowledge base. Most of ES shells present a way to introduce new data to the system by converting it into rules or schemes, but this procedure is often very complex and not intuitive for the expert who is going to use and update the software (Thomas, 1989; Sheikhtaheri et al., 2014).

Another problem could be represented by the introduction of new and more complex rules: the probability of conflicts between existing rules raises, thus making the update of the knowledge base a long and difficult process (Thomas, 1989). Therefore, ES updates are very complex and possibly expensive in case of necessary knowledge, requirement and/or functionality changes (Thomas, 1989; Yetisen et al., 2018).

Recently, due to the technological and computational advances, ES advantages are more perceptible, especially when these systems are considered as decision supports for the human experts (Oyediji et al., 2019). As it was pointed out by Singla et al. (2014), expert systems have increased their reliability, reduced errors and costs, organized multiple expertise in a unique knowledge base, thus simplifying the access and consultation of expert knowledge.

However, the ESs are not devoid of problems and traditional expert system applications have clearly shown their limits over the years. In the following are reported the issues highlighted by Duan et al. (2005).

- **Knowledge Bottleneck.** Knowledge gathering can be complicated, due to the heterogeneity of sources and experts unclear explanation of their decision process.
- **Performance brittleness.** Every ES is limited to its operating area of expertise, thus it performs poorly outside of it.
- **Availability and Software Distribution.** Since every ES is likely used as a stand-alone system, it can be difficult to (1) provide it at the place and time where it is needed and (2) install and update it over time without the help of an expert.
- **Communication between distributed applications.** There is a lack of common protocols for ES knowledge, resulting in a difficult co-operation and dynamic information sharing.

However, Duan et al. (2005) see the advent of Internet as an aid to improve ES through new and accessible technologies for both experts and non-experts.

Nevertheless, even if there are numerous examples of web-based ESs, they are relatively small and since just some of them have been analyzed thoroughly it is still difficult to assess the methodologies and challenges behind the design of these expert systems and thus understand their advantages and issues (Duan et al., 2005).

4. Validation

Having explored the main characteristics of ESs and knowing the deriving advantages and issues coming from the application of these systems in different fields, we identified the validation process as an important step for ES evaluation (Asabere, 2012; Maylawati et al., 2018). Having heterogeneous fields of application, each ES has to be validated accordingly to the necessities identified by the problem at hand, trying to achieve the concept of trust explained in Section 2.2.

In the early 90s, Grogono, Preece, Shinghal, and Suen (1993) identified the main evaluation steps an ES should undertake: *structural* and *functional* testing, after having established the minimum and desired performance levels; *laboratory* and *field* evaluations for real-life assessment.

Having these concepts in mind, we analyzed the validation methods present in the ES literature and especially the ones concerning the applications described in Section 5. We noticed some similarities between different applications and that the validation, if present, may involve one or more of the steps described in the following.

The *usability* and *functional testing*, as well as the one performed only by the experts, usually involve the administration of questionnaires. For example, Duan et al. (2005) analyze 3 web-based ESs, whose evaluation process consisted in satisfaction, effectiveness and usability studies performed by students and fishermen.

Other evaluation steps may involve the testing on *simulated*, *controlled* or *real data*, sometimes complemented by the *comparison with standard diagnostic techniques* and the *application of machine learning methods* to assess the ES performance. Usually, the performance is represented by *external measures* like accuracy, sensitivity and specificity.

Table 1 provides a summary of the best reported performances (obtained through the application of external measures) for each field of application presented in Section 5.

Notice that the analyzed urinary system disease applications do not refer to any kind of validation process of this type.

As previously mentioned, the validation may involve different steps. Braido et al. (2018) perform an *in silico* validation with non-specified IT techniques, a pilot validation by feeding the system with diagnosed patients' data and a clinical validation on new patients, controlled by the experts.

Instead, Lin et al. (2010) and Kadhimi et al. (2011) use real-data collected from hospital patients and apply external measures to test their ES performances, while Prasad, Prasad, and Sagar (2011) focus on the introduction of different algorithms in their ES, in order to evaluate their accuracy, validity, reliability and effectiveness in detecting asthma, adding a comparison with experts' diagnoses in terms of specificity and sensitivity.

Even Seto et al. (2012) use real-data collected on 2 groups of 50 patients, one receiving standard care without their ES aid and one provided with their system, to detect heart failures. However, their only aim is to verify if their ES would provide effective health benefits to their patients.

Another scenario is proposed by Mirchevska, Luštrek, and Gams

Table 1

Best reported performance for each field of application in respect to the validation undertaken by the state-of-the-art works reported in Section 5.

Application	Best reported performance
Heart related	(Valenza et al., 2008) specificity and sensitivity about 98%; (Lin et al., 2010) accuracy about 94%
Fall detection & elderly problems	(Rescio et al., 2013) specificity and sensitivity about 95% and 98%; (Siddiqui et al., 2020) accuracy about 96%
Neurology & chronic pains	(Montaña et al., 2018) accuracy about 92% - 94%
Pulmonary diseases	(Badnjevic et al., 2018) accuracy about 97%
Urinary system diseases	None
Others	(Sevani et al., 2020) precision and recall about 0.66 and 1

(2014), who design an expert controlled experiment to detect different types of fall through their ES, testing its robustness, accuracy and adaptation capabilities.

This kind of validation, even though could represent a good way to access the system trust (Vourgidis et al., 2018; Josefiok et al., 2017), is unfortunately overlooked in most of the works we have reported in our review. As it was highlighted by Josefiok et al. (2017), most of ESs are validated on accuracy and just a few are evaluated in practice tests (an observation confirmed by our analyses, reported and discussed in Table 2 and Section 6, respectively). Also, many works present little to no information about the validation of their systems (Section 6), this lack being an obstacle to understand the efficacy of the systems and their impact on real-life application (Asabere, 2012; Maylawati et al., 2018). For example, in (Abu-Naser & Aldahdooh, 2016) the only information available about the system validation is that some students were involved in the process and that they considered the system adequate and comfortable to interact with, making their validation non-reproducible.

5. Applications

As stated in the previous sections, we focus our attention on studies concerning the application of ESs in the healthcare domain. In fact, developed countries continue to experience significant problems in providing healthcare services and telemonitoring could be identified as a promising tool with the potential to relieve some of the burden on the healthcare system, allowing patients to take care of themselves and enabling more efficient clinical care, for example through automatic alerts at the first sign of deterioration of patients' health (Seto et al., 2012).

Monitoring physiological information in real-life conditions is particularly useful to manage chronic disorders or health-care problems (Chen, 2011). Thus, ESs have been used in a great variety of medical applications and we analyze some studies on heart (Chen, 2011; Das et al., 2013; Kaur & Kaur, 2014; Seto et al., 2012; Valenza, Lanatà, Ferro, & Scilingo, 2008; Lin et al., 2010; Oleshchuk & Fensli, 2011; Thai, Minh, & Phung, 2017; Serkova et al., 2017; Ali et al., 2019), fall detection and elderly problems (Rescio et al., 2013; Mirchevska et al., 2014; de la Concepción, Morillo, García, & González-Abril, 2017; Yacchirema, de Puga, Palau, & Esteve, 2018; Almarashdeh et al., 2018; El-Mashharawi, Alshawwa, Elkahout, & Abu-Naser, 2019; Abu-Naser, 2015; Siddiqui et al., 2020; Marlinda et al., 2020), neurology and chronic pains (Kadhim et al., 2011; Abu-Naser & Aldahdooh, 2016; Santra et al., 2019; El Agha, Jarghon, & Abu-Naser, 2017; Ijla & Sabri, 2020; Josefiok et al., 2017; Bobić et al., 2019; Montaña, Campos-Roca, & Pérez, 2018), pulmonary (Prasad et al., 2011; Fu, Ayyagari, & Colquitt, 2009; Badnjevic, Gurbeta, & Custovic, 2018; El Kahout, Alshawwa, El-Mashharawi, & Abu-Naser, 2019) and urinary system (Abu-Naser & Shaath, 2016; Mirmozaffari, 2019; Rivera Jr et al., 2019) diseases applications.

Table 2 summarizes the state-of-the-art works analyzed in the following subsections, highlighting their main characteristics and the validation approach (if any).

Notice that we include in our analysis some miscellaneous works on medical expert systems (Oyedede et al., 2019; Oyelade, Obiniyi, Junaidu, & Adewuyi, 2018; Sevani, Sagala, & Kristiani, 2020; Naser & Al-Bayed, 2016; Nunes, Pinheiro, & Pequeno, 2009; Latif et al., 2017; Soldani et al., 2017; Chen, 2011; Rathore, Ahmad, Paul, Wan, & Zhang, 2016; Chowdhury et al., 2020) and that we discuss a couple of examples about methods and techniques that may represent a support and improvement to the reported MESs (Xue, Lu, & Chen, 2019; Tian et al., 2019; Sanchiz, Amadiou, Paubel, & Chevalier, 2020; Hussain et al., 2018).

5.1. Heart-related applications

Many ES applications have been developed for heart-related diseases recognition and management, wanting a continuous monitoring of the

Table 2

Expert system applications in the medical field analyzed state-of-the-art works.

Reference	Application	Main characteristics	Validation
Abu-Naser and Aldahdooh (2016)	Lower back pain	Rule-based, experts are the source of knowledge	Tested by experts
Abu-Naser and Shaath (2016)	Urination problems	Rule-based, decision tree algorithm	None
Ali et al. (2019)	Heart failure prediction	Stacked SVM models	Test on different experiments and against other machine learning models, providing a set of external measures
Almarashdeh et al. (2018)	Elderly monitoring	Wearable sensors, DB server and GUI	Testing on software and hardware
Badnjevic et al. (2018)	Asthma and chronic obstructive pulmonary disease	Inference engine, artificial neural network, fuzzy logic	Tested in 2 steps on real-data
Bobić et al. (2019)	Bradykinesia	Rule-based, wearable sensors, control units and signal processing algorithm	Tested by experts
Chen (2011)	Human pulse monitoring system	Sensors, detection algorithm and web-based interface	Comparison with standard diagnostic techniques
Chen et al. (2017)	Remote diagnosis of psychological and physiological diseases	Design of the 5G-Csys system	Prototype system
Chowdhury et al. (2020)	Optical camera communication for eHealth solutions	Optical camera communication and IoT devices, 5G architecture	None
Das et al. (2013)	Hypertension diagnosis	Fuzzy rules	Evaluation on prediction accuracy
de la Concepción et al. (2017)	Activity recognition and fall detection	Smartphone, data collection, classification	Tested on real-data; accuracy
El Agha et al. (2017)	Polymyalgia rheumatica	Rule-based, friendly interface	None
El Kahout et al. (2019)	Silicosis diagnosis and treatment	SL5 Object language, knowledge base	Evaluated by medical experts
El-Mashharawi et al. (2019)	Arthritis diagnosis	Knowledge base, inference engine, user interface	Tested by young experts
Fu et al. (2009)	Pulmonary diseases	Wearable sensor, environment sensor and central controller: rule-based ES and UI	None
Honglun et al. (2013)	Fall detection	Wearable sensor, algorithm for fall detection	Testing by volunteers, evaluation on accuracy and success rate
Ijla and Sabri (2020)	Chronic chest pain	Knowledge representation, user interface	None
Josefiok et al. (2017)	Neurology	Research questions, characteristics of a work on ES	None
Kadhim et al. (2011)	Back pain	Fuzzy rules, knowledge base, inference engine, user interface	Testing based on clinical data, evaluation of the system accuracy
Kaur and Kaur (2014)	Hypertension diagnosis	Knowledge acquisition and fuzzy rules	None

(continued on next page)

Table 2 (continued)

Reference	Application	Main characteristics	Validation
Latif et al. (2017)	eHealth solutions	5G architecture, wearable devices, IoT, AI	None
Lin et al. (2010)	Atrial fibrillation detection	Wearable device, detection algorithm	Clinical tests, plus evaluation on sensitivity, specificity and positive prediction rate
Marlinda et al. (2020)	Elderly health monitoring	Certainty factor method	None
Mirchevska et al. (2014)	Fall detection	Domain knowledge and machine learning, rule-based classifier	Design of a test scenario to investigate the generality and robustness of the classifiers (accuracy and F-measure)
Mirmozaffari (2019)	Nephrolithiasis diagnosis and treatment	Knowledge base design	None
Montaña et al. (2018)	Parkinson's disease detection	Knowledge base, feature extraction	Tested on a narrow dataset, external measures
Naser and Al-Bayed (2016)	Video-game addiction	Knowledge base, user interface, inference engine and rule-based	None
Nunes et al. (2009)	Psychological disorders	Knowledge representation based on rules and probabilities, machine inference	None
Oleshchuk and Fensli (2011)	Patient monitoring (focus on heart related applications)	Wearable sensors, data collection, 5G architecture	None
Oyedeji et al. (2019)	Students' typical pathologies	Knowledge base, rule base, inference engine and web application	Validation to check the performance, the functionality and validity of the software
Oyelade et al. (2018)	Breast cancer	GUI, NL Parser and Lexicon database, inference model, rule base	None
Prasad et al. (2011)	Asthma	Knowledge base, decision tree algorithms	Application with real data; accuracy, validity, reliability and effectiveness testing
Rathore et al. (2016)	Citizens vitals monitoring and Big Data analysis	Medical IoT sensors, big architecture to connect different infrastructures	Tested on 3 UCI datasets, a heart rate dataset and the ILPD dataset
Rescio et al. (2013)	Fall detection-elderly monitoring	Wearable sensor, data collection, feature extraction in the time-domain	Evaluation on sensitivity and specificity in controlled conditions: tested on actors
Rivera Jr et al. (2019)	Kidney diseases diagnosis	Mobile technologies, knowledge base definition, fuzzy logic and decision trees	User evaluation through questionnaire
Santra et al. (2019)	Lower back pain	Knowledge representation, inference engine, conflict management	Tested on patient records and compared with expert diagnoses; accuracy, precision, recall

Table 2 (continued)

Reference	Application	Main characteristics	Validation
Serkova et al. (2017)	Heart disease destabilisation assessment	Fuzzy knowledge base	Tested on real-data
Seto et al. (2012)	Heart failure	Rule-based, knowledge acquisition from experts	Experimental and usability testing
Sevani et al. (2020)	Diabetes mellitus patients' healthy lifestyle	Knowledge base, inference engine, PHP and MySQL	Expert evaluation, precision and recall computation, usability evaluation
Siddiqui et al. (2020)	Arthritis diagnosis	Fuzzy variables, inference engine	Tested on simulated data
Skacel and Soukup (2019)	Risk of atherosclerotic plaques	Rule-based, web application	None
Soldani et al. (2017)	Tele-surgery and eHealth solutions	Mobile technologies, 5G architecture	None
Thai et al. (2017)	Heart disease diagnosis	IoT, noise removal, feature extraction	Tested on real-data
Valenza et al. (2008)	Cardiac arrhythmia detection	Wearable system, feature extraction and classification	Specificity and sensitivity computation
Yacchirema et al. (2018)	Fall detection	IoT, wearable devices, big data and alert management	Tested on controlled experiment; external measures computation

patients' physiological information in real-life situations.

For example, [Chen \(2011\)](#) presents a novel web-based remote human pulse monitoring system with intelligent data analysis based on physiological sensors. The system is composed by piezoelectric sensor for sensing pulse, signal pre-processing circuits for amplification and filtering, an embedded system platform for pulse collection and wireless transmission. Also, a web based interface is designed to observe immediate pulse signal, the real-time QRS detection algorithm computes the pulse rate and the system exploits cosine similarity measure to explore abnormal pulse sequences.

To verify that the implemented system can correctly measure pulse signals, they compare the pulse signal displayed on the web interface with the value on an oscilloscope.

Another of the main topics of expert systems concerning heart-related diseases is the diagnosis of hypertension.

The study conducted by [Das et al. \(2013\)](#) presents a fuzzy expert system and (feed-forward) neuro fuzzy based system (NFS) for the diagnosis of hypertension. The comparison among these 2 systems shows the effectiveness and accuracy of using neuro fuzzy based system over FES. They conclude that NFS represents a useful method for medical diagnostic task of finding hypertension risk factor.

Exploiting the fuzzy logic of these peculiar kind of systems, [Kaur and Kaur \(2014\)](#) developed their own FES to determine the risk of hypertension. This FES receives as inputs 7 variables that could allow the identification of the risk of hypertension and outputs if the risk is present or not.

Another promising strategy to improve the outcomes of cardiac problems is telemonitoring. By making remote patient monitoring possible, the clinicians can intervene early if there is evidence of clinical deterioration. Therefore, the purpose of the work conducted by [Seto et al. \(2012\)](#) is to develop a rule-based ES for a heart failure mobile phone-based telemonitoring system. First they create a rule set based on expert interviews. The rule set is then evaluated as part of a 6-month randomized controlled trial of the telemonitoring system. The results from the post-trial interviews indicate that the expert system led to positive self-care behavior changes.

Following the concept of early and quick detection of heart-diseases, it is also important to obtain it in case of electrocardiogram (ECG) arrhythmia, especially for the treatment of patients in the intensive care unit. Valenza et al. (2008) built up a wearable system for the recognition of multiple cardiac arrhythmia. In particular, pattern recognition of QRS complex is performed from binding minimal architecture of neural network as Kohonen self organizing map (Kohonen, 1990). Experimental results were validated obtaining specificity and sensitivity up to 98%.

Instead, Lin et al. (2010) propose a wireless ECG device equipped with a built-in automatic warning ES. The acquired ECG signals are transmitted to mobile devices through Bluetooth, and then processed by the ES. After performing clinical tests the accuracy achieved by the system was approximately 94%, with high sensitivity, specificity and positive prediction rate. This study demonstrates that the proposed intelligent system is capable of accurately detect atrial fibrillation episodes and instantaneously alert both the user and the healthcare personnel, facilitating early medical intervention.

Wanting to exploit new communication technologies, Oleshchuk and Fensli (2011) suggest the use of 5G networks to have a reliable and fast remote monitoring of patients, while ensuring system security. They describe a use case related to heart failure and arrhythmia detection exploiting wearable sensors and pull/push mechanism to allow a bi-directional communication between patients and experts.

Exploiting the advantages of IoT technologies, Thai et al. (2017) propose an expert system able to manage ECG signals, by removing the present noise with a combination of the sequential recursive, discrete wavelet transform and Fisher linear discriminant algorithms. They also provide a feature extraction mechanism to capture the heart disease information and finally evaluate the system by using a publicly available real-data dataset.

Another MES application is for coronary heart disease destabilization assessment (Serkova et al., 2017), which is modeled with fuzzy knowledge bases, improvable at any stage.

The data of interest are processed and integrated into the knowledge base and the system tested on a patient group and on a control one. Through this evaluation, the researchers verified that the level of soluble vascular adhesion molecules could be used as a criterion to assess heart disease destabilization.

A final realization of expert system for heart failure prediction is developed by Ali et al. (2019) and exploits machine learning techniques, i.e., stacks 2 support vector machines (SVMs) in order to firstly select the optimal feature subset and secondly obtain its predictions.

In fact, the system takes in input a public heart disease dataset and pass it firstly to a L_1 -norm and then to a L_2 -norm regularized linear SVM model; afterwards, it merges the obtained (2) sets of hyperparameters in order to create a hybrid grid. This grid is passed to a hybrid grid search algorithm, which provides the optimal feature subset and the optimized predictive model.

The authors evaluate the proposed strategy on 3 experiments with a holdout validation and by providing accuracy, sensitivity, specificity and Matthews correlation coefficient values. They also compare their results with other machine learning techniques, showing the system efficiency and reliability.

5.2. Fall detection and elderly problems

A good number of ES applications deals with fall detection and monitoring of elderly people. In fact, an ES could aid patients, especially elderly people, who need long-term attention for their physical conditions, such as chronic diseases and falls, with ubiquitous surveillance and remote management of recorded health data and immediate assistance (Honglun, Meimei, & Minghui, 2013).

Rescio et al. (2013) propose a fall detector through a wearable triaxial microelectromechanical system (MEMS) accelerometer. The wearable device can operate in streaming (raw data are sent via ZigBee

to an external computing platform for data analysis with a 10 Hz frequency) or in standalone modality. In order to analyze the wave forms along each axis in the presence of falls and other kind of events (activities of daily living), a data collection has been defined in controlled (simulated) conditions. In these conditions, a high performance in terms of sensitivity and specificity was obtained using the 20% of the dataset for training. Misclassifications are for falls presenting slow dynamics or falls with partial recovery.

Instead, Mirchevska et al. (2014) evaluate the combination of domain knowledge and machine learning (CDKML) methods for classifier generation in a fall detection application. Firstly, prior domain knowledge is enriched with relevant patterns obtained by machine learning techniques to create an initial classifier. Secondly, genetic algorithms refine the classifier. Finally, the classifier is adapted online on the basis of user feedback through a Markov decision process.

Tests show that the classifiers developed after the first 2 phases are already more reliable and robust than ML classifiers built from limited examples from the domain of interest. The adapted classifier is thus inclined to reduce the number of misclassifications during sitting on a low chair at the cost of not detecting certain falls after which the user lands sitting on the ground. The method is suitable for domains that have limited obtainable training data and available domain knowledge.

A more recent work on fall detection (de la Concepción et al., 2017) uses smartphones as monitoring devices. The authors apply Ameva (Gonzalez-Abril, Cuberos, Velasco, & Ortega, 2009) to obtain discrete variables and use a subject-based approach for classification. They perform an activity recognition not limited to fall detection, and achieve the 95% of accuracy on the dataset collected from 30 volunteers.

Yacchirema et al. (2018) also propose a IoT based system for fall detection of elderly people, adding to their system not only smart devices, but also embedded wireless sensors. A wearable device is in fact provided to the elderly person and it assesses the smart IoT gateway present in the elderly person home through a wireless communication network. The gateway transforms and analyzes the big data, basing its decisions on the data and machine learning model present in the cloud services to which it connects. Using its knowledge base, it handles the alerts and, in case a fall is detected, notifies the elderly person caregivers.

Finally, the authors evaluate their system on a controlled experiment, simulating the falls, and they compute accuracy, precision, recall and gain, noticing a sufficiently high detection rate.

Moving to the prevention and control of chronic diseases for elderly patients, the system developed by Almarashdeh et al. (2018) focuses on elderly people instability, immobility, intellectual impairment and incontinence, while making the physicians aware of the patients and their problems, supporting them when necessary.

The system is composed by smart bracelets with embed medical sensors, by a database (DB) server that uses wireless technology to receive data from the bracelets and the physicians' personal device assistants that are connected to the DB server through Internet infrastructure communications. The physician can view the patient medical history on the interface application and receive an alert message indicating the patient status if in need of attention.

Instead, El-Mashharawi et al. (2019) propose an ES for arthritis diagnosis, especially meant to assist young and expert orthopedists as well as their patients. The system follows the typical ES modeling, having a knowledge base composed by the specific literature and expert works on the topic; relying on Simpler Level 5 (SL5) Object language (Abu-Naser, 2015) to create facts, rules and objects to be used in the inference engine; and being provided with a user interface. Notice that the ES was positively evaluated by orthopedic students.

On the same topic, Siddiqui et al. (2020) use an adaptive hierarchical Mamdani FES for arthritis detection and early diagnosis. The system receives in input a series of fuzzy variables, representative of the arthritis symptoms, processes them through a fuzzifier, whose output is passed to the inference engine. The inference engine is able to make its

decision, basing its outcome on the transformed inputted variables and the experts' rules provided by the knowledge base. Finally, after the defuzzification, the FES outputs the diagnosis.

The authors tested the system on simulated data and obtained an average accuracy of the 95.6% on 4 diagnostic conditions.

As a final MES intended for monitoring elderly general health on a daily basis, we report the work of [Marlinda et al. \(2020\)](#). The authors propose a theoretical model based on the certainty factor method, which they would like to exploit in order to measure the confidence on a specific decision, being in presence of uncertain data. However, they point out that it is a debated method and that it may require multiple data processing. Thus, it is an open issue for further research.

5.3. Neurology and chronic pains

In the field of neurology and chronic pains, a rapid and early diagnosis significantly improves patient prognosis and allows the application of appropriate measures. Therefore, the main objective of the ESs is to offer the best possible support to medical staff and patients when dealing with these problems.

[Kadhim et al. \(2011\)](#) apply a fuzzy expert system to allow back pain diagnosis. They interestingly introduce the concept of *metaknowledge*, defined as the information retaining the experts' approach in exploiting their knowledge to address a specific problem. Therefore, the knowledge relative to the domain of interest may be general, being it acquired through literature and human experts, but also meta.

The developed FES models the clinical observation symptoms into fuzzy rules and its outcome allows the diagnosis of back pain by the physician and also to send advice to the patients.

In fact, the system is structured to take as inputs the users' personal information, the symptoms and other necessary clinical information, taking into account the patients' clinical history. Through the inference engine and the knowledge base, the system outputs diagnosis and advice.

Another ES proposed for lower back pain (LBP) diagnosis is the one developed by [Abu-Naser and AlDahdooh \(2016\)](#) by asking yes/no questions. The system is implemented using SL5 Object language and, at the end of the processing stage, the ES gives a diagnosis and some recommendations for the user. The main sources of the knowledge for this ES are physicians and specialized websites for LBP. Currently the ES has 10 rules. A group of medicine students voluntarily accepted to participate in testing the expert system. They were satisfied with the performance of the proposed ES and they were comfortable using the system.

A more recent study of LBP is the one conducted by [Santra et al. \(2019\)](#), who concentrate on the definition of an efficient knowledge representation, a diagnostic inference engine and the introduction of a Bayesian network to resolve conflicts when facing the possibility of having more diseases associated to the inputted data.

The resulting MES was tested on 30 LBP patients and its outcomes compared with the expert diagnoses. The system efficiency was assessed through accuracy, precision and recall external measures. The MES responded closely to the medical expert diagnoses, even though the clinical investigations were not taken into consideration.

Several research groups have developed expert systems to support people suffering from chronic pain. For example, an ES was designed by [El Agha et al. \(2017\)](#) for diagnosing Polymyalgia Rheumatic problems and it recommends the proper treatment by asking yes/no questions. The main goal of this work is to help people in self-diagnose with the aid of a specialized ES. This specific system has the advantages of being easy to use and of having a user friendly interface.

A more recent ES application is the one proposed by [Ijla and Sabri \(2020\)](#) for chronic chest pain diagnosis and treatment. The authors focus on 8 types of chest pains and heavily rely on the user interface to have a clinical picture from which detect the problem and suggest its solution. The work gives a detailed explanation on the knowledge representation, specifying that the information were collected from physicians and

specialized websites, to finally be translated into CLIPS syntax.

For what concerns the neurological applications, [Josefiok et al. \(2017\)](#) discuss some works regarding the application of ESs as support for medical analyses in neurology. An interesting contribution is represented by the research questions used to make a literature review: (1) existing approaches of ESs in the domain of interest; (2) application of these systems in real life; (3) limits and future developments. The authors also highlight the lack of standards, which could represent an obstacle to compare different systems.

Finally, the authors identify 3 main characteristics a work on ES application to neurology should emphasize: the foundation of the knowledge base, its development and the system validation.

A more specific application is the one developed by [Bobić et al. \(2019\)](#), who model a decision support ES for objective evaluation and automatic scoring of bradykinesia in repetitive finger-tapping movements for patients with idiopathic Parkinson's disease and atypical parkinsonism.

The ES consists of simple rules designed to match universally defined criteria that have been evaluated in clinical practice. The proposed system is composed by 2 inertial sensors connected to control units that acquire and wirelessly transmit sensor data to a remote computer. The methodology is based on automatized and repeatable signal-processing techniques.

Finally, the accuracy of the system was calculated based on the reference scores provided by 2 neurologists. The introduced solution results convenient being repeatable, easy to implement and potentially able to provide good assistance in clinical practice.

On the other hand, it is still missing a graphical interface that shows the numerical results, scores and statistical analysis.

Another application is proposed by [Montaña et al. \(2018\)](#), who focus on detecting Parkinson's disease at an early stage, in order to ameliorate the patients' quality of life. They identify the knowledge base as the core element of an ES and thus provide a novel method based on voice onset time segments to extract temporal and spatial features, populating their database. They obtain a high accuracy on a dataset collected from patients at a lower disease stage.

5.4. Pulmonary diseases

Chronic pulmonary diseases, such as asthma, often require a regular schedule of treatment and on-going reassessment for the duration of a patient's life.

The research group of [Prasad et al. \(2011\)](#) develop an expert system, applying machine learning methods to diagnose asthma. Especially, auto-associative memory neural network resulted the best technique to diagnose this kind of diseases.

The knowledge base is constructed exploiting physicians' experts knowledge and the application is modeled on real-data publicly available to have more clinical relevance. Finally, the ES is evaluated through questionnaires and clinical tests, including also the accuracy computation of the machine learning models.

[Fu et al. \(2009\)](#) implement a pulmonary disease management system with on-body and near-body sensors for real-time monitoring of patients' daily activities. The ES is composed of 3 different parts: (1) *body monitor* to check the pulmonary activity, (2) *environment monitor* to assess the environment conditions and (3) *central controller* constituted by the core of the ES and data processing tools.

Instead, [Badnjevic et al. \(2018\)](#) want to detect asthma and chronic obstructive pulmonary disease, by proposing an expert diagnostic system that could be integrated into a healthcare system.

The proposed ES takes a preliminary decision taking as the input a symptom based questionnaire. If the resulting classification detects the presence of a possible pulmonary disease, the system recommends a set of diagnostic tests, whose results are subsequently inputted to the system until a robust decision is reached. The decision process is modeled with an artificial neural network based on fuzzy logic.

The preliminary decision process was evaluated on 2735 samples and checked by medical experts, while the entire ES was evaluated in real-time on 1650 patients, obtaining an overall accuracy greater than 97%.

El Kahlout et al. (2019) develop an ES specifically intended for the diagnosis and treatment of silicosis.

The system is modeled using SL5 Object language, its knowledge base relies on the information present on websites covering the topic of interest and it is composed by 40 rules.

Finally, the ES was evaluated by medical experts, which found out that the system results were near to the actual silicosis diagnoses.

5.5. Urinary system diseases

Other researches have concentrated their efforts in the detection and treatment of different diseases affecting the urinary system.

Abu-Naser and Shaath (2016) report in details the implementation and design of an ES able to diagnose urination diseases through a decision tree. It uses SL5 Object language which, we remind, is a production system executing a rule-based program.

On a correlated topic, Mirmozaffari (2019) describes an ES environment, in order to support the physicians in the diagnosis and treatment of patients affected by nephrolithiasis. The authors focus their efforts in designing an adequate knowledge base, dividing it in 3 sections that pertain to (1) the required variables for program execution; (2) the rules governing the symptoms detection and the system recommendations; (3) the system response to the given inputs.

An interesting observation reported by the authors is about the validity of the inputs, which they declare to not be able to detect.

Concerning the kidney diseases diagnosis, Rivera Jr et al. (2019) develop a mobile expert system based on fuzzy logic and build its knowledge base using the state-of-the-art studies on this topic as well as information obtained directly from field experts.

The symptoms filtering and the diagnostic testing were executed through the application of decision tree and fuzzy logic algorithms. The mobile application complementing the ES was validated by different system users (nephrologists, patients and mobile developers), which had to answer a set of questions on a Likert scale (Robinson, 2014). The questionnaire was designed following the ISO 9126 Software Quality Characteristics standard.

5.6. Other medical applications

In this section we report miscellaneous ES applications in the medical domain.

Oyedeji et al. (2019) present the development of a system that is able to accurately diagnose some common diseases such as malaria and typhoid. They consulted a group of medical practitioners for knowledge acquisition. The user interacts by entering necessary information in English and the system responds using a backward chaining (deductive reasoning) process to get an answer to the questions asked by the user. In particular, the waterfall design model has been used for the software development. A validation step is used to check the performances and all the functionalities of the software.

A peculiar type of ES uses a natural language parsing approach. The proposed formalized input model (Oyelade et al., 2018) provides support for generating acceptable medical terms or tokens as input for medical diagnostic systems, in particular it is tested on a breast cancer diagnosis. The main aim of this paper is to ensure that patients are allowed to enter their input into the system in an open ended pattern and then formalizes them model mapping the input into acceptable list of symptoms. Their approach uses a patient input box (GUI), natural language parsers, crafted rules sets, a domain based lexicon and an inference engine.

Sevani et al. (2020) develop an integrated MES for the promotion of diabetes mellitus patients' healthy lifestyle. The system should be able to

identify the problem a patient is facing and query its knowledge base to answer it. The knowledge base is constituted by expert knowledge and research on the topic, while the inference engine is provided with backward chaining methods.

Notice that the ES is built using PHP and MySQL and tested by system developers and by 86 users. The obtained results comply with the expert opinions and both interface and functionality were evaluated as satisfactory. Precision (0.658) and recall (1) were computed as external measures for ulterior system validation, however the authors suggest a more complex evaluation of their proposal.

A great number of other research groups decided to apply the theory of ESs to psychological disorders and addictions (Naser & Al-Bayed, 2016; Nunes et al., 2009; Alshawwa, Elkahout, El-Mashharawi, & Abu-Naser, 2019). Despite the predominance of drug-related addictions, there is a number of behaviors that could be considered as potentially addictive. Naser and Al-Bayed (2016) introduce an expert system to help users in getting the correct diagnosis of video game addictions by asking questions that require yes/no answers.

According to the World Health Organization (WHO), health is a state of complete physical, mental and social well-being. In the domain of psychological disorders, abnormal behavior is difficult to define because of subjectivity.

Nunes et al. (2009) observe that an ES for modeling decision making support in this specific field must take into account several variables, including cultural, biological, psycho-social issues, quality of information, signs and symptoms. This is called multi-criteria decision-making process. The methodology is composed by 3 main stages: (1) the identification of goals, (2) the evaluation phase and (3) the sensitivity and robustness analyses to check whether changes in the parameters of the model affect the final outcome.

As a final set of application examples, we describe 5 works concerning the exploitation of 5G technologies in order to have more reliable, secure and efficient remote healthcare systems.

Latif et al. (2017) highlight the importance of having a patient-centric healthcare system, where governance, legal, commercial and communication aspects revolve around the patients. They describe in details the opportunities that could arise using 5G infrastructures, being them an enhancement to previously studied services: real-time, lifeline and highly-reliable communications, broadcast-like services, broadband access from potentially everywhere, higher user mobility and exploitation of IoT.

The solution for tele-surgery proposed by Soldani et al. (2017) relies on the aforementioned characteristics to have a reliable and low-latency communication between experts and patients. They focus on the idea of removing cyber-sickness when an operation is performed remotely, having as key factors a short and flexible transmission time interval, the management of transmitted packets, low energy consumption, the efficient control of wireless robots and edge computing centers.

Chen, Yang, Hao, Mao, and Hwang (2017) propose an innovative eHealth system called 5G Cognitive System (5G-Cs). The system allows the cognition of haptic movements and thus it could be used for tele-surgery. 5G-Cs is also able to perform recognition of humans' affective states in order to treat patients' mental illnesses. Therefore, this particular system works on 2 different parts: a resource cognitive engine and a data cognitive engine. The first one is meant to learn network contexts and aims at ultra-low latency and ultra-high reliability for cognitive applications. On the other hand, the second one is based on the analysis of healthcare big data and is used to handle a patient's health status physiologically and psychologically.

Another major topic is addressed by Rathore et al. (2016), who focus their attention on processing big data collected through a variety of sensors placed on a great number of subjects, performing real-time actions in critical situations. In order to tackle those 2 challenges, they present a real-time medical emergency response system, whose task is to continuously monitor the subjects' health efficiently by using IoT medical sensors placed on the human body. The aim of this system is to

keep track of the vital signals of individuals thanks to an intelligent building, which puts in contact hospitals, emergency services, first aid and police stations and eventually allows these actors to intervene.

Finally, Chowdhury et al. (2020) propose a system based on 5G communication and the usage of optical camera communication devices. In their architecture there is always a direct communication between different networks, being them continuously connected to the monitoring devices and the patients' sensors.

5.7. Support methods

In this section, we give some hints on possible support methods for the previously analyzed applications and strategies.

Notice that a good number of the cited works rely on the information collected from the literature, from interviewing experts and from specialized websites, thus obtaining knowledge bases that could contain a set of uncertain and complex concepts (Nohria, 2015).

Aligning the resulting ontology, may represent a way to clarify the medical concepts and terms collected from different sources. In fact, Xue et al. (2019) propose the application of a non-dominated sorting genetic algorithm (NSGA)-III to match a biomedical ontology and demonstrate the effectiveness of their proposal in determining high-quality ontology alignments.

The right importance should also be given to a properly structured and accessible UI, since not all the practitioners may have an extensive technical knowledge to operate a complex system. In (Sanchiz et al., 2020), for example, the focus is on how to help older adults perform Internet searches even with the decline of their fluid abilities. The developed system is designed to offer the proper support to reformulate queries more rapidly and to elaborate more flexible searching strategies. In (Hussain et al., 2018) the goal of having personalized and accessible systems is even more clear, since the designed model aims to deliver an adaptive user interface created with an A-UI/UX-A tool that can be tweaked looking at user evaluation and experience.

We conclude this brief section, pointing out that another major improvement of the described ESs could be represented by the introduction of new and more efficient machine learning techniques, especially when exploited to provide better input data to the system itself.

For example, Tian et al. (2019) propose an enhanced convolutional neural denoising network, that could represent an aid for medical systems using images for disease diagnosis. The authors prove that their method allows an easier training of the network compared to other state-of-the-art works and obtain cleaner images.

6. Discussion

The problem of medical diagnosis is an issue complicated by different factors (Kadhim et al., 2011), especially because this matter contemplates the acquisition and interpretation of multiple kinds of information: patient's symptoms can be obtained from his/her anamnesis, signs, laboratory test results, ultrasonic results, X-ray findings and so on (Dev et al., 2016). Interpreting this information in a correct way is a delicate matter, the interpretation depending solely on the expertise and perception of the practitioner (Sikchi et al., 2013). In fact, even with many years of experience, a physician could still draw the wrong conclusion. Some pathologies, such as urinary diseases (Abu-Naser & Shaath, 2016) or lower back pain problems (Abu-Naser & Aldahdooh, 2016), are still mis-diagnosed, translating into erroneous therapies that do not help the patient. Other conditions, such as motor symptoms of patients with neurological disorders (Bobić et al., 2019) or chronic obstructive disease (Braido et al., 2018), can be particularly difficult to be differentiated and recognized properly even by the experts. This is where medical expert systems, with also the support of wearable sensors and new communication technologies, can provide important support to physicians and doctors (Section 5).

They can prove to be, for example, important tools to handle

imprecise and ambiguous information (Kadhim et al., 2011), or a point of major improvement in patients' lives, especially thinking about elderly or chronically ill people (Silva et al., 2015).

In our review we noticed that most of the heart related applications use FESs (Kaur & Kaur, 2014; Serkova et al., 2017) complemented with wireless and/or wearable devices (Chen, 2011; Seto et al., 2012; Valenza et al., 2008; Lin et al., 2010), which are extensively used to detect falls (Rescio et al., 2013; de la Concepción et al., 2017; Yacchirema et al., 2018). In the neurology and chronic pains domain, the ESs are designed in order to give out recommendations, especially when dealing with LBP (Kadhim et al., 2011; Abu-Naser & Aldahdooh, 2016; El Agha et al., 2017). The MESs concerning the pulmonary disease diagnosis and treatment are particularly heterogeneous and focus on providing the systems with efficient machine learning techniques (Prasad et al., 2011), designing a well-structured rule set (El Kahlout et al., 2019) and find wearable solutions (Fu et al., 2009). Moreover, the urinary system diseases present different applications that are concerned with the introduction of machine learning methods (Abu-Naser & Shaath, 2016), well-defined knowledge bases (Mirmozaffari, 2019) and mobile applications (Rivera Jr et al., 2019).

Notice that the information collection to populate the knowledge bases follows a general criterion (Section 5): searching the literature and specialized websites on the medical problem of interest, complementing the obtained information by interviewing medical experts.

In the miscellaneous reported medical applications, we have found some pointers for future and more thorough analyses, especially: the usage of natural language processing when dealing with the ES inputs (Oyelade et al., 2018), the development of MESs through other languages like PHP and MySQL (Sevani et al., 2020), the consideration of user subjectivity (Nunes et al., 2009), the exploitation of 5G technologies (Latif et al., 2017; Chen et al., 2017; Chowdhury et al., 2020) and Big Data (Chen et al., 2017).

These observations also justify the idea of introducing an analysis on the supporting techniques that may improve the ES design and data collection (Section 5.7) in future works.

Even though the MESs have demonstrated their efficacy, we encountered a peculiar shortage in properly tested and evaluated projects (Section 5). Looking at Fig. 4 and 5 it is easy to notice how many works paid little to no attention to the validation step of their project, only presenting their design and functioning rather than completing its analysis with exhaustive testing. Concentrating on Fig. 4, we can observe how more than half of the analyzed papers performed only 1 or even no type of tests on their system (of the 43 selected works, 15 with just 1 evaluation method performed and 14 without validation) and just in 3 papers we can find information about more than 2 performed strategies for validation.

Another information we can extract from Fig. 5 is that the external measures strategy for validation is the most utilized among the works we analyzed, confirming the observations made by Josefiok et al. (2017) and reported in Section 4. The testing on real data is performed by 10 contributions, but the works that did not apply any evaluation strategy remain the most numerous in respect to the more developed studies. Overall, the missing or limited validation step is an important lack of information in these papers, since it is crucial to understand if a theorized system can also be used effectively in practice, and it is one of the things that should be further considered when developing this kind of solutions in the future.

Notice that in our review we wanted to give a brief but broad overview of the different applications an ES may be involved with when facing the medical domain.

Therefore, we made a selection based on the criteria described in the introduction (Section 1), obtaining 43 heterogeneous works written between 2008 and 2020 and presenting at least one of the main components an ES should be provided with (Section 2) and we observed that most of the proposed ESs represent an improvement, or at least an aid, to the classical diagnosis and treatment procedures given by the experts.

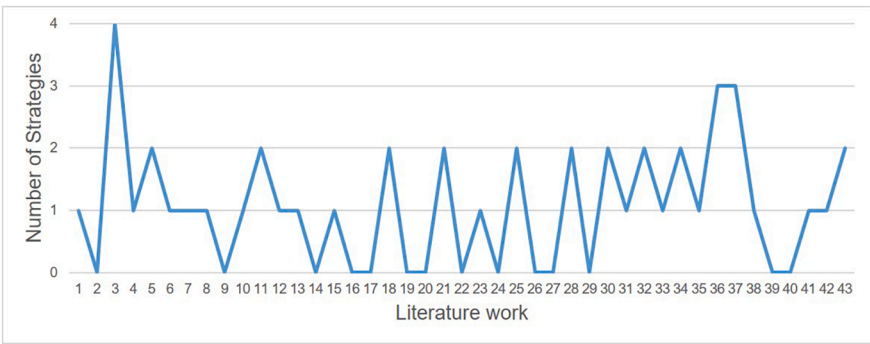


Fig. 4. Number of strategies used for validation by each of the 43 analyzed works.

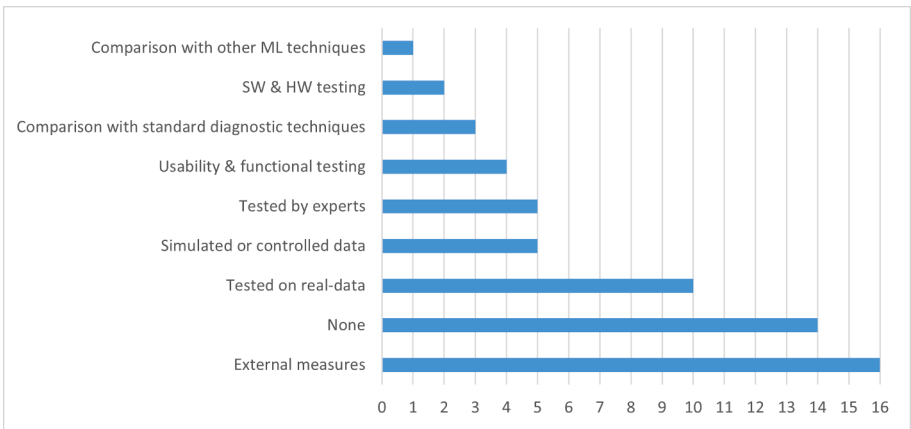


Fig. 5. Strategies applied for validation sorted by the overall number of occurrences. The acronyms correspond to Machine Learning (ML), Software (SW), Hardware (HW).

The conducted researches and analyses are based on our expertise in the computer science field and on the results provided by the referenced work authors, thus our evaluation is biased towards the technical aspects of the proposals more than on their medical evaluation.

In the future, it would be enriching providing the analyzed state-of-the-art contributions to medical experts in order to have a more balanced evaluation of the systems.

For completeness, we report an infographic (Fig. 6) about the standard medical approaches for the diagnosis and treatment of the analyzed medical macro areas and we refer the readers to the following papers in order to obtain more information on some of the described topics from a

medical perspective:

- Alzheimer (Weller & Budson, 2018).
- Arthritis (Shepherd & Mullins, 2019).
- Asthma (Busse, 2011).
- Gait disorders (Snijders, Van De Warrenburg, Giladi, & Bloem, 2007).
- Heart failure (Inamdar & Inamdar, 2016).
- Lower back pain (Maher, Underwood, & Buchbinder, 2017).
- Parkinson's disease (Savitt, Dawson, & Dawson, 2006).
- Polymyalgia rheumatica (Soubrier, Dubost, & Ristori, 2006).

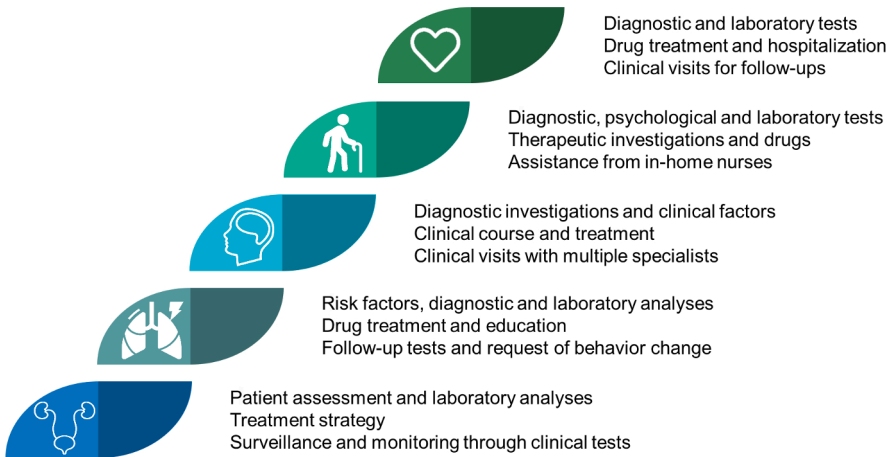


Fig. 6. A graphical summary of the common medical approaches for the treatment and diagnosis of some of the analyzed diseases.

- Pulmonary diseases (Hopkinson, Molyneux, Pink, & Harrisingh, 2019).
- Urinary tract infections (Tony Mazzulli, 2012).

We also give a graphical summary (Fig. 7) on the power of expert systems applied in the medical domain. Notice that the main advantages deriving from the usage of MESs, compared with the brief report on the standard approaches of Fig. 6, are (1) the system accessibility for each kind of user, being the system modeled for patients, caregivers and physicians alike (Almarashdeh et al., 2018); (2) the ability to provide an aid for monitoring, management, diagnosis and treatment of a peculiar disease, with a unique system that is able to centralize the diagnostic and personal information and to take decisions based on the data (Singla et al., 2014); (3) a decrease in economical and system costs, if the MES is well-defined, in respect to the in-home assistance or the frequent clinical visits (Sam, Srinidhi, Niveditha, Amudha, & Usha, 2020); (4) the possibility of having a constant communication between different users and thus ask for assistance in any moment (Latif et al., 2017); and (5) the flexibility, which allows an out-of-the-hospital care, while remaining efficient and reliable (Maylawati et al., 2018).

7. Conclusions

In this work we presented a literature review on expert systems, especially for applications in the medical field.

Our main contribution was to highlight the importance of a validation step in order to obtain an ES that can be trusted by its users (Vourgidis et al., 2018), being them physicians or patients alike; to present a broad overview of the different ES realizations by searching the state-of-the-art for heterogeneous and significant works (Section 5); and to point out some future directions for the medical expert system development and analyses (Section 6).

Firstly, we presented a detailed description of the components that constitute an ES (Section 2), then we analyzed the specific characteristics of FESs (Section 2.1) and WESs (Section 2.3); subsequently we discussed the advantages and issues of these methodologies (Section 3), followed by a detailed discussion of the validation process (Section 4); finally, we reported and compared some state-of-the-art works concerning medical applications (Section 5) and discussed some critical points deriving from our analyses (Section 6). Generally, a medical expert system seems to represent a useful decision support system in daily clinical practice and thus to be able to improve the effectiveness and quality of healthcare (Seto et al., 2012).

Even though ESs make the domain experts' knowledge and decision processes more accessible and relieve experts from certain tasks (Asabere, 2012), they could be affected by some issues like the difficulty of

system updating and the restrictive way of storing knowledge (Sheikhtaheri et al., 2014).

Therefore, these systems should be validated in order to ensure their reliability and dependability, using functional and structural testing before the deployment (Grogono et al., 1993; Duan et al., 2005). However, we noticed that the ESs performance is usually evaluated by computing only external measures and that a few of them are validated in real-world settings (Section 6).

Through the comparison of medical applications we noticed that almost half of the research works includes the development of a web application or a graphical user interface to ease web-page search and access to results and to make the system more user friendly (Section 5). Moreover, the majority of the ESs is rule-based and only a few of them are based on fuzzy rules. These latter are particularly suitable for handling complex problems such as the diagnosis of specific diseases and we hypothesize that in the future FESs will be integrated in medical applications. We also point out that a future analysis may profit from the introduction of ulterior systems based on different mathematical solutions that are able to model uncertainty and vagueness (Ilczuk et al., 2005; Sevastianov et al., 2012; Gambhir et al., 2016).

Concerning the structure of the knowledge base, we observed great variability in the description of knowledge acquisition and representation. In particular, just a limited number of papers explicitly mentioned the process of knowledge collection, which is usually performed by searching the state-of-the-art and by directly asking experts to provide information (Section 6).

In our literature search, we noticed that a great number of works focuses on heart diseases related applications (Section 5.1). They use different approaches, being the authors more interested in fuzzy logic, wearable sensors flexibility, knowledge acquisition or in a combination of different technologies and models.

We also encountered a large amount of works focusing on neurological diseases and chronic pain (Section 5.3). Their main aims are giving recommendations for treatment procedures or supporting the medical caregivers for disease prevention and patients' monitoring. Furthermore, elderly problems (Section 5.2) have become a growing topic of interest in the last years and wearable expert systems are the most suitable to address this task. In particular, in combination with embedded sensors, specific algorithms were developed to detect instability, immobility, intellectual impairment and incontinence.

Exploiting the described flexibility of ESs and combining their models, their medical applications should gain more trust from their users, being them physicians and/or patients. A trust funded on system reliability, efficacy, validity and acceptance.

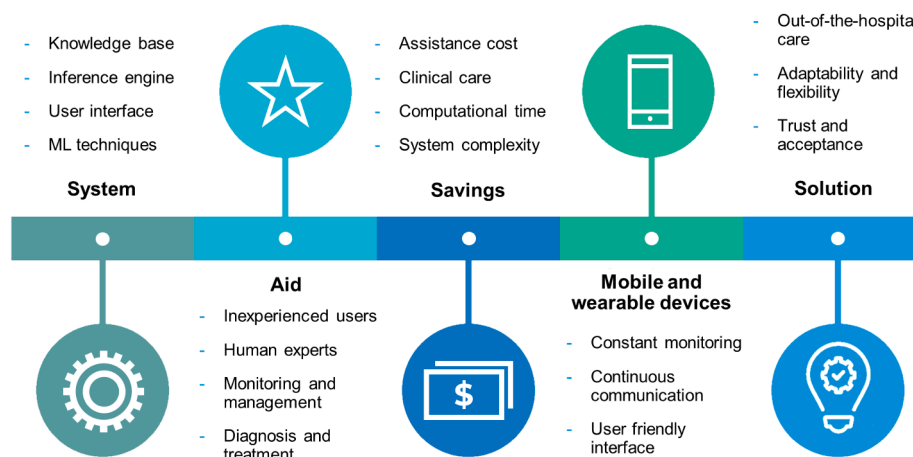


Fig. 7. A graphical summary of the medical expert systems features.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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